

Phase 3: Key Insights and Future Directions

A comprehensive insights report by Yinglu Deng (Merck Sharp & Dohme, Rahway, NJ, USA) and Yang Xu (Longseemed Medicine Inc., Guangdong, China)

1. Overview

We developed and evaluated a machine learning framework for the early prediction of Alzheimer's Disease and Related Dementias (AD/ADRD) using spontaneous speech.

- **Final Model Architecture:** A multi-task model built on a pre-trained OpenAI Whisper encoder, featuring a primary classification head for diagnosis (Healthy Control, MCI, or ADRD) and **a novel auxiliary temporal segmentation head designed to identify speech disfluencies for enhanced interpretability.**
- **Datasets Used:** The models were trained on 9 datasets (n=2,058) and evaluated on 4 distinct English-language speech corpora using a leave-one-out policy: the Wisconsin Longitudinal Study (WLS; n=825), the DementiaBank Pitt Corpus (PITT; n=549), the VAS corpus (n=100), and the Delaware corpus (n=81). We also utilized an expert-labeled fluency dataset (mispeech/speechocean762) to train and evaluate the disfluency analysis module.
- **Novel Speech Fluency Biomarkers:** We propose three novel biomarkers for speech analysis. First is the collection of disfluencies identified from ASR text. Second is the word token probability, which indicates the difficulty of identifying a word from the speaker's speech. The third is the transferred fluency score from the speechocean762 dataset. All three fluency scores show statistically significant decreases from healthy controls to ADRD patients.
- **Future Directions:** Given the success of our novel biomarkers and auxiliary labels, we suggest that with limited datasets, creating more detailed and accurate temporal labels can help improve model performance and generalizability. The data collection standard operating procedures (SOPs) are also critical, as we suspect this represents the primary factor introducing bias between datasets.

2. Performance

- **Model Performance Overview.** Our model achieved a **1st place** ranking during the modeling stage, demonstrating its outstanding performance on the competition dataset. To further assess its generalization capabilities, we conducted a cross-corpora evaluation using a leave-one-out methodology. The results, as detailed in Table 1, indicate that the integration of an auxiliary temporal segmentation head improved the model's generation performance across all corpora except for the Spanish corpus. We used a fine-tuned state-of-the-art Qwen2.5-Omni multimodal LLM to provide the fluency score labels. The specific log-loss results are presented below.

Table 1 benchmarks our 1st place Phase II solution(Whisper encoder + FC layer). This model was enhanced using Class Activation Maps, word confidence, and fluency score as auxiliary labels for temporal segmentation.

Auxiliary Labels 	WLS (ENG, n=825)	PITT (ENG, n=549)	VAS (ENG, n=100)	DELAWARE (ENG, n=81)	IVANOVA(ESP, n=360)
(N/A) Phase II model	0.859	1.049	1.10	2.09	1.04
Word Confidence	0.898	0.880	0.967	2.05	1.07
Fluency Score	0.839	0.869	0.956	1.87	1.11

- **Groups with Lower Performance and Potential Improvements:** The model demonstrated the most pronounced prediction errors on the Spanish dataset and smaller corpora, namely the Delaware and VAS datasets, as indicated by their substantially higher log-loss scores. To diagnose these performance disparities, we performed a decomposition analysis for each audio sample, visualizing the outcomes in Figure 1a. This examination uncovered a notable distributional divergence between the Spanish dataset, along with segments of the English dataset, and the general population. We ascribe this discrepancy to a distinct data collection methodology utilized for these particular datasets. The logloss of each corpus at competition test set (Figure 1b) also shows that outliers (ivanova and vas) have higher value than inliers.

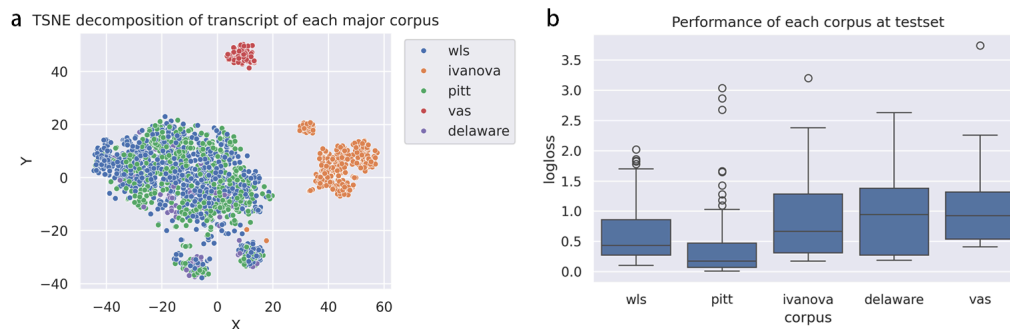


Figure 1, Analysis of performance bias among each group, **a.** the decomposition of each audio's language feature, ivanova(Spanish dataset) and vas(English with special) both have a notable distributional divergence. **b.** the performance of each corpus at competition test set, The outliers we discovered at Figure 1a, have higher logloss than inliers.

• Key Limitations for Clinical Use:

Besides the limited generalizability between datasets, we also face following challenge:

- **Limited Data:** The lack of large, diverse, and multimodal longitudinal datasets.
- **Proof-of-Concept Status:** The models are for research purposes only and require rigorous, prospective clinical validation before they can be considered for diagnostic use.

3. Methodology

- **Auxiliary Learning Improves Performance and Interpretability:** The implementation of a multi-task architecture with an auxiliary head for predicting temporal features proved to be a highly effective strategy, consistently improving performance over baseline classifiers. This approach not only acts as a form of regularization but, more critically, enhances model interpretability by generating "fluency maps" and Class Activation Mapping (CAM) scores to visualize the specific speech events influencing predictions, directly addressing the "black box" problem in medical AI.

Figure 2 depicts the comprehensive information available to users for decision-making. Figure 2a shows basic audio information, while Figure 2b displays the Class Activation Map, highlighting regions important to the model's decision. Figures 2c and 2d present features completely independent of the model that can serve as external validation tools for checking model predictions. In this case, the two areas highlighted by the model's CAM were also captured by our fluency features: at 6-7 seconds, there is both a low fluency score (Figure 2c) and a detected block (Figure 2d); at 18-22 seconds, another long block is present (Figure 2d). Manual verification confirmed expression issues in each flagged zone, as shown in Figure 2e.

This provides a practical workflow where users can first understand the model's logic by inspecting these scores, then efficiently verify the output using independent features and listening only to flagged audio segments before accepting the prediction.

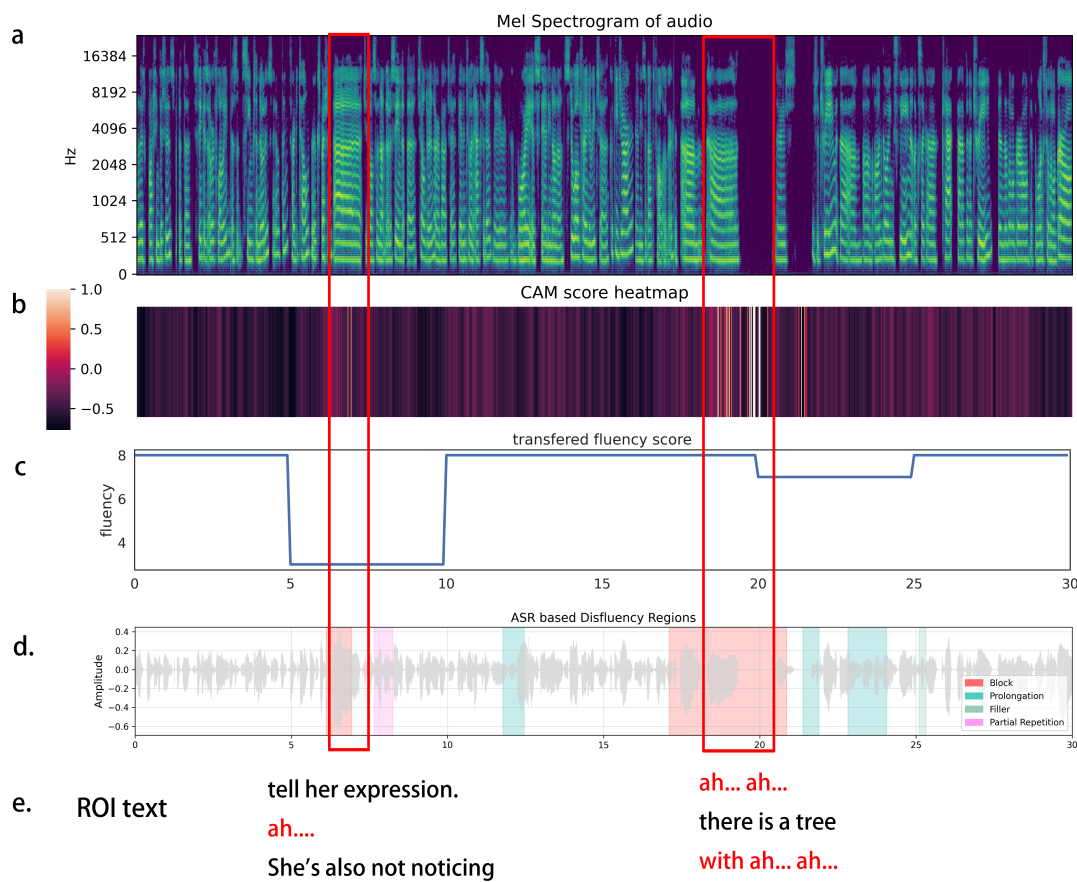


Figure 2, the visualization of the model Interpretability. **a.** the Mel Spectrogram. **b.** The Class activation map, highlight the region where model's decision based on. **c.** the transferred fluency score, which is irrelevant with classification model. **d.** the ASR based disfluency feature, which is also irrelevant with classification model.

- New Fluency Biomarkers for Speech Analysis:** We propose three novel biomarkers for speech analysis. First, **disfluency patterns:** the frequency of pauses, repetitions, and false starts was a highly significant differentiator between diagnostic groups ($p=6.6 \times 10^{-9}$). This feature provides a direct, quantifiable measure of the word-finding difficulties (anomia) and impaired cognitive planning that are clinical hallmarks of the disease. Second, **ASR word confidence:** the average confidence score from the ASR model was an extremely potent predictor ($p=1.98 \times 10^{-14}$). It functions as an automated proxy for speech clarity, likely capturing subtle articulatory imprecision and vocal changes that may be among the earliest signs of neurodegeneration. Third, **transfer-learned fluency:** the fluency score derived from a model trained on non-native speech was also highly significant ($p=3.37 \times 10^{-19}$). This demonstrates that the fundamental acoustic and temporal characteristics of disrupted speech provide a robust and transferable signal of impairment, even when the underlying causes differ.

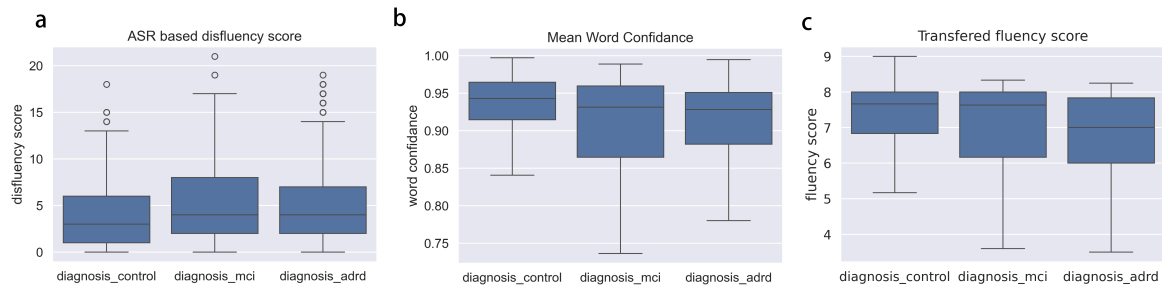


Figure 3. a. The disfluency score identified by the disfluency patterns of ASR text. b. The ASR word confidence. c. the fluency score predicted by the LLM trained on an expert fluency dataset.

- **Large Model Pretraining with Extensive Datasets Improves Performance:** Generally, larger pretrained models with more extensive datasets perform better, and this principle holds true for audio tasks. In our Phase II benchmark, Whisper, which was trained on the most extensive pretraining dataset, outperformed other methods. In Phase III, we further evaluated the current state-of-the-art Qwen2.5-Omni multimodal LLM. Results showed that balanced accuracy improved from 0.582 to 0.665.
- **Multi-modal Models Enhance Performance:** By training a BERT model on textual features, we discovered that in a small subset of cases where speech was fluent but semantically nonsensical, the BERT model predicted a high probability of disease. While this did not improve overall performance scores, we consider this an interesting finding that warrants further investigation, as it suggests that semantic analysis can capture cognitive impairments that acoustic measures might miss.

4. Data

- **Effectiveness of Cross-Domain Transfer Learning:** The successful application of a fluency model trained on non-native speaker data to the AD/ADRD cohort was a significant finding. It demonstrates that it is possible to "borrow" statistical power from larger, related datasets in other domains to overcome data scarcity in a specific clinical domain. This validates the strategy of using transfer learning to build powerful feature extractors when target data is limited.
- **Recommendations for Future Data Collection:** To meaningfully advance the field, strategic data collection is the highest priority. We propose the following three core recommendations:
 - **Establish Standardized Data Collection and Quality Control Protocols:** To eliminate bias at the source and enhance the reproducibility and reliability of research findings, we strongly recommend the community collaboratively develop and adopt a Standard Operating Procedure (SOP) for data collection. Currently, valuable resources like DementiaBank contain data from diverse sources with varying collection methods, which introduces unnecessary variables and potential biases into model training. By implementing such a protocol, we can minimize technical heterogeneity and more accurately isolate the genuine speech signals associated with the disease.
 - **Champion Demographic and Linguistic Diversity:** Future efforts must actively recruit from underrepresented populations and build parallel corpora across many languages. This is the only way to develop equitable tools and determine which speech markers of dementia are universal versus language-specific.
 - **Leverage LLMs for Automated, Privacy-Preserving Assessment Tools:**
We recommend using Large Language Models (LLMs), to develop a new generation of automated assessment tools. These applications, deployable on smartphones or personal devices, can achieve **Automated Testing and Evaluation** and **On-Device Anonymization and Privacy Preservation**. It enables scalable, longitudinal, and low-cost monitoring of an individual's functional health while maximally protecting participant privacy.

5. Contributing Factors

The study confirmed that computationally engineered features serve as powerful, objective biomarkers that align with real-world clinical manifestations of AD/ADRD. Through Class Activation Map (CAM) analysis of our machine learning model, we found that the timestamps highlighted by the model were highly correlated with disfluency regions (Figure 2). Combined analysis of transcript decomposition (Figure 1a) and the text-based machine learning model revealed that context is also a critical factor influencing the model's predictions and can introduce bias in certain datasets.

6. Future Directions

- **Recommended Next Steps for Advancing Prediction:**

- **Create and Apply Universal Data Collection and Processing Pipeline:** Although DementiaBank has implemented careful attention to dataset collection and management, enabling easy access and analysis across different data sources, most Non-Protocol Data (such as WLS and PITT corpus) share only the Cookie Theft task, while other assessment methods remain different. Corpora showing significant performance decay, such as VAS and Delaware, utilize different protocols—voice-activated systems versus DementiaBank Protocol, respectively. Therefore, we recommend that future data collection should follow standardized protocols to minimize bias and improve cross-dataset compatibility.
- **Expert Annotation for High-Precision Labels:** While we demonstrated in our methodology that algorithmically generated labels for features like fluency can improve model performance, the accuracy of these labels remains a concern. We recommend a focused effort on obtaining expert annotations to create high-precision labels for these diagnostically relevant Regions of Interest (ROI) to further boost model performance.
- **Saliency-Based ROI Selection:** Develop models that can automatically scan long speech samples to identify and focus analysis on the most diagnostically relevant segments (ROI), which could improve both accuracy and efficiency.
- **Privacy-Preserving AI for Large-Scale Screening:** For real-world deployment, privacy is paramount. Future work should focus on benchmarking voice conversion tools to anonymize speaker identity and implementing federated learning frameworks, where models are trained on a user's local device without the raw speech data ever being centralized.

- **Key Limiting Factors in the Field:**

- **Data Scarcity and Heterogeneity:** The lack of large, diverse, and multimodal longitudinal datasets remains the single greatest obstacle to progress.
- **Clinical and Ethical Hurdles:** Beyond algorithmic development, significant challenges remain. For real-world impact, these tools must be responsibly integrated into clinical workflows as Clinical Decision Support (CDS) systems, not standalone diagnostics. This requires addressing issues of data bias to ensure equity, building trust with patients and clinicians, and navigating the complex regulatory landscape for Software as a Medical Device (SaMD) with bodies like the FDA, including adherence to new AI-specific guidelines and reporting standards.

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